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Temperature controller, substrate processing apparatus, and pressure control method

Abstract

A temperature controller includes a supply line configured to supply a heat medium to at least one control target, a return line configured to return the heat medium from control target, a pump configured to discharge the heat medium to the supply line, a pressure sensor configured to detect a pressure in the supply line disposed immediately after the pump, and a controller. The controller is configured to control the temperature controller to control a discharge pressure of the pump based on a difference between the pressure detected by the pressure sensor and a set value of the discharge pressure of the pump.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application No. 2021-053733 filed on Mar. 26, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a temperature controller, a substrate processing apparatus, and a pressure control method.

BACKGROUND

(3) In a substrate processing apparatus, temperature control is performed by circulating brine that is a heat medium through a flow path in a substrate support using a pump. For example, it is proposed to vaporize a temperature control medium in a cavity in a substrate support to control a pressure in the cavity to a target pressure (see U.S. Patent Application Publication No. 2012/0016508). Further, it is proposed to prepare a reference table for a relationship between a wafer temperature and a coolant pressure in advance and control the coolant pressure based on wafer temperature drift using the reference table at the time of measurement (see U.S. Patent Application Publication No. 2002/0048311). Further, it is proposed to control a flow rate of a coolant in a pump or a temperature control system including a high temperature fluid source, a low temperature fluid

source, and a three-way valve using an operating frequency of an inverter (see U.S. Patent Application Publication No. 2020/0051839 and No. 2019/0355598).

SUMMARY

(4) The present disclosure provides a temperature controller, a substrate processing apparatus, and a pressure control method capable of improving temperature controllability for a control target.

(5) One aspect of the present disclosure provides a temperature controller includes a supply line configured to supply a heat medium to at least one control target, a return line configured to return the heat medium from control target, a pump configured to discharge the heat medium to the supply line, a pressure sensor configured to detect a pressure in the supply line disposed immediately after the pump, and a controller, wherein the controller is configured to control the temperature controller to control a discharge pressure of the pump based on a difference between the pressure detected by the pressure sensor and a set value of the discharge pressure of the pump.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The objects and features of the present disclosure will become apparent from the following description of embodiments, given in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a schematic cross-sectional view showing an example of a substrate processing apparatus according to an embodiment of the present disclosure;

(3) FIG. 2 shows an example of a temperature controller according to the embodiment;

(4) FIG. 3 is a flowchart showing an example of a pressure control process according to the embodiment;

(5) FIG. 4 shows an example of temperature characteristics in a test example;

(6) FIG. 5 shows an example of temperature characteristics in a comparative example;

(7) FIG. 6 shows an example of a temperature controller in a first modified example;

(8) FIG. 7 shows an example of a control target in a second modified example; and

(9) FIG. 8 shows an example of a temperature controller in the second modified example.

DETAILED DESCRIPTION

(10) Hereinafter, embodiments of a temperature controller, a substrate processing apparatus, and a temperature control method of the present disclosure will be described in detail with reference to the accompanying drawings. The following embodiments are not intended to limit the present disclosure.

(11) In circulation of a heat medium, for example, a flow rate of a heat medium on a high temperature side and a flow rate of a heat medium on a low temperature side may be switched for a substrate support as a temperature control target, or the number of substrate processing apparatuses as a temperature control target may increase or decrease. In that case, a pressure in a line through which the heat medium is supplied varies considerably due to interference between valves, or the like, so that circulation controllability deteriorates. Accordingly, an operation of a pump becomes unstable, which may deteriorate temperature controllability for the temperature control target. In the case of performing flow rate feedback control or performing control while fixing an inverter frequency of the pump for circulation control of the pump, it is difficult to suppress deterioration of circulation controllability. Further, in the case of switching a line on a return side using a check valve, a valve on a high temperature side and a valve on a low temperature side may be opened at the same time without controlling a pressure difference, which may also result in deterioration of the temperature controllability. In this case, alarm stop occurs due to an insufficient liquid amount or an excessive liquid amount caused by deviation of a liquid level of a tank on a high temperature side and a liquid level of a tank on a low temperature side, or the temperature controllability deteriorates due to excessive heat loss caused by liquid movement through a communication line

between the tank on the high temperature side and the tank on the low temperature side. Hence, it is expected to improve the temperature controllability for the control target by controlling a pressure on a discharge side of the pump to be constant.

(12) (Configuration of Substrate Processing Apparatus 1)

(13) FIG. 1 is a schematic cross-sectional view showing an example of a substrate processing apparatus according to an embodiment of the present disclosure. The substrate processing apparatus 1 is, e.g., a plasma etching apparatus including parallel plate electrodes. The substrate processing apparatus 1 includes an apparatus main body 10 and a controller 11. The apparatus main body 10 has a processing chamber 12 made of a material such as aluminum or the like and formed in a substantially cylindrical shape, for example. The processing chamber 12 has an anodically oxidized inner wall surface. Further, the processing chamber 12 is frame grounded.

(14) A substantially cylindrical support portion 14 made of an insulating material such as quartz or the like is disposed at a bottom portion of the processing chamber 12. In the processing chamber 12, the support portion 14 extends vertically (e.g., toward an upper electrode 30) from the bottom portion of the processing chamber 12.

(15) A substrate support PD is disposed in the processing chamber 12. The substrate support PD is supported by the support portion 14. The substrate support PD holds a wafer W on an upper surface thereof. The wafer W is an example of a temperature control target. The substrate support PD has an electrostatic chuck ESC and a lower electrode LE. The lower electrode LE is made of a metal material such as aluminum or the like and has a substantially disc shape. The electrostatic chuck ESC is disposed on the lower electrode LE. The lower electrode LE is an example of a heat exchange member for exchanging heat with the temperature control target.

(16) The electrostatic chuck ESC has a structure in which an electrode EL that is a conductive film is embedded between two insulating layers or between two insulating sheets. A DC power supply 17 is electrically connected to the electrode EL through a switch SW. The electrostatic chuck ESC attracts the wafer W on the upper surface thereof by an electrostatic force such as a Coulomb force generated by a DC voltage supplied from the DC power supply 17. Accordingly, the electrostatic chuck ESC can hold the wafer W.

(17) A heat transfer gas such as He gas or the like is supplied to the electrostatic chuck ESC through a line 19. The heat transfer gas supplied through the line 19 is supplied to a gap between the electrostatic chuck ESC and the wafer W. By adjusting a pressure of the heat transfer gas supplied to the gap between the electrostatic chuck ESC and the wafer W, thermal conductivity between the electrostatic chuck ESC and the wafer W can be adjusted.

(18) Further, a heater HT that is a heating element is disposed in the electrostatic chuck ESC. A heater power supply HP is connected to the heater HT. By supplying a power from the heater power supply HP to the heater HT, the wafer W on the electrostatic chuck ESC can be heated through the electrostatic chuck ESC. A temperature of the wafer W placed on the electrostatic chuck ESC is adjusted by the lower electrode LE and the heater HT. The heater HT may be disposed between the electrostatic chuck ESC and the lower electrode LE.

(19) An edge ring ER is disposed around the electrostatic chuck ESC to surround an edge of the wafer W and the electrostatic chuck ESC. The edge ring ER may also be referred to as “focus ring.” The edge ring ER can improve in-plane uniformity of processing on the wafer W. The edge ring ER is made of a material that is appropriately selected depending on a material of a film to be etched, such as quartz or the like.

(20) A flow path 15 through which a heat medium that is an insulating fluid such as Galden (Registered Trademark) or the like flows is formed in the lower electrode LE. The heat medium may also be referred to as “brine.” A temperature controller 20 is connected to the flow path 15 through lines 16a and 16b. The temperature controller 20 controls a temperature of the heat medium flowing in the flow path 15 of the lower electrode LE. The heat medium whose temperature is controlled by the temperature controller 20 is supplied into the flow path 15 of the

lower electrode LE through the line **16a**. The heat medium flowing in the flow path **15** is returned to the temperature controller **20** through the line **16b**.

(21) The temperature controller **20** circulates the heat medium flowing in the flow path **15** of the lower electrode LE. Further, the temperature controller **20** mixes a heat medium having a first temperature or a heat medium having a second temperature with the circulating heat medium and supplies the mixed heat medium into the flow path **15** of the lower electrode LE. The temperature of the lower electrode LE is controlled to a set temperature by supplying the mixture of the heat medium having the first temperature or the heat medium having the second temperature with the circulating heat medium into the flow path **15** of the lower electrode LE. The first temperature is, e.g., a temperature of room temperature or higher, and the second temperature is, e.g., a temperature of 0° C. or lower. Hereinafter, the heat medium having the first temperature will be referred to as “first heat medium” and the heat medium having the second temperature will be referred to as “second heat medium.” The first heat medium and the second heat medium are fluids made of the same material and having different temperatures. The temperature controller **20** and the controller **11** are examples of a heat medium controller.

(22) A power supply line **69** for supplying a radio frequency (RF) power to the lower electrode LE is electrically connected to a bottom surface of the lower electrode LE. The power supply line **69** is made of a metal. Although it is not illustrated in FIG. **1**, lifter pins for transferring the wafer W on the electrostatic chuck ESC, a driving mechanism thereof, and the like are disposed in the space between the lower electrode LE and the bottom portion of the processing chamber **12**.

(23) A first RF power supply **64** is connected to the power supply line **69** through a matching device **68**. The first RF power supply **64** generates an RF power for attracting ions into the wafer W, i.e., an RF bias power, having a frequency of 400 kHz to 40.68 MHz, e.g., 13.56 MHz. The matching device **68** is a circuit for matching an output impedance of the first RF power supply **64** with an input impedance on a load (lower electrode LE) side. The RF bias power generated by the first RF power supply **64** is supplied to the lower electrode LE through the matching device **68** and the power supply line **69**.

(24) The upper electrode **30** is disposed above the substrate support PD to face the substrate support PD. The lower electrode LE and the upper electrode **30** are substantially parallel to each other. Plasma is generated in a space between the upper electrode **30** and the lower electrode LE, and plasma processing such as etching or the like is performed on the wafer W held on the upper surface of the electrostatic chuck ESC by the generated plasma. The space between the upper electrode **30** and the lower electrode LE serves as a processing space PS.

(25) The upper electrode **30** is supported at an upper portion of the processing chamber **12** via an insulating shielding member **32** made of, e.g., quartz or the like. The upper electrode **30** has an electrode plate **34** and an electrode support **36**. A bottom surface of the electrode plate **34** faces the processing space PS. A plurality of gas injection holes **34a** are formed in the electrode plate **34**. The electrode plate **34** is made of a material containing silicon, for example.

(26) The electrode support **36** is made of a conductive material such as aluminum or the like, and detachably supports the electrode plate **34** from the top. The electrode support **36** may have a water-cooling structure (not shown). A diffusion space **36a** is formed in the electrode support **36**. A plurality of gas holes **36b** communicating with the gas injection holes **34a** of the electrode plate **34** extend downward (toward the substrate support PD) from the diffusion space **36a**. The electrode support **36** is provided with a gas inlet port **36c** for introducing a processing gas into the diffusion space **36a**, and a line **38** is connected to the gas inlet port **36c**.

(27) A gas source group **40** is connected to the line **38** through a valve group **42** and a flow rate controller group **44**. The gas source group **40** has a plurality of gas sources. The valve group **42** includes a plurality of valves, and the flow rate controller group **44** includes a plurality of flow rate controllers such as a mass flow controller and the like. Each of the gas sources in the gas source groups **40** is connected to the line **38** through the corresponding valve in the valve group **42** and the

corresponding flow rate controller in the flow rate controller group **44**.

(28) Accordingly, the apparatus main body **10** can supply a processing gas from one or multiple gas sources selected from the gas source group **40** to the diffusion space **36a** in the electrode support **36** at an individually controlled flow rate. The processing gas supplied to the diffusion space **36a** is diffused in the diffusion space **36a** and supplied in a shower pattern into the processing space PS through the gas holes **36b** and the gas injection holes **34a**.

(29) A second RF power supply **62** is connected to the electrode support **36** through a matching device **66**. The second RF power supply **62** generates an RF power for plasma generation having a frequency of 27 MHz to 100 MHz, e.g., 60 MHz. The matching device **66** is a circuit for matching an output impedance of the second RF power supply **62** with the input impedance on the load (upper electrode **30**) side. The RF power generated by the second RF power supply **62** is supplied to the upper electrode **30** through the matching device **66**. The second RF power supply **62** may be connected to the lower electrode LE through the matching device **66**.

(30) A deposition shield **46** whose surface is made of aluminum coated with Y.sub.2O.sub.3, quartz, or the like is detachably attached to an inner wall surface of the processing chamber **12** and an outer surface of the support portion **14**. The deposition shield **46** can prevent etching by-products (deposits) from being adhered to the processing chamber **12** and the support portion **14**.

(31) Between an outer wall of the support portion **14** and an inner sidewall of the treatment chamber **12**, an exhaust plate **48** made of aluminum coated with Y.sub.2O.sub.3, quartz, or the like is disposed on the bottom side (where the support portion **14** is disposed) of the processing chamber **12**. An exhaust port **12e** is disposed below the exhaust plate **48**. An exhaust device **50** is connected to the exhaust port **12e** through an exhaust line **52**.

(32) The exhaust device **50** has a vacuum pump such as a turbo molecular pump or the like, and can reduce a pressure in an inner space of the processing chamber **12** to a desired vacuum level. An opening **12g** for loading/unloading the wafer W is disposed on a sidewall of the processing chamber **12**. The opening **12g** can be opened and closed by a gate valve **54**.

(33) A controller **11** has a processor, a memory, and an input/output interface. The memory stores a program executed by the processor and a recipe including conditions for each processing. The processor executes the program read from the memory, and controls individual components of the apparatus main body **10** through the input/output interface based on the recipe stored in the memory to perform predetermined processing such as etching or the like on the wafer W. The controller **11** is an example of a control device.

(34) (Configuration of Temperature Controller **20**)

(35) FIG. 2 shows an example of a temperature controller according to the present embodiment. The temperature controller **20** includes a circulator **200**, a first temperature controller **220**, and a second temperature controller **240**. For example, the circulator **200** is installed on the same level as the level where the processing chamber **12** is installed, and the first temperature controller **220** and the second temperature controller **240** are installed on a level lower than the level where the circulator **200** is installed.

(36) In the circulator **200**, an outlet side of a valve **201** is connected to the line **16a**. Further, in the circulator **200**, a pump **202** for circulating a heat medium flowing in the flow path **15** of the lower electrode LE is connected to the line **16b**. The line **16b** on the outlet side of the pump **202** is connected to an inlet side of the valve **201** through a check valve **206** and a line **207** at a connection position A. Since a pressure of the line **207** is lower than a pressure of the line **16b** on the outlet side of the pump **202** during an operation of the pump **202**, the check valve **206** is opened. Therefore, the heat medium circulates in the order of the pump **202**, the line **16b**, the check valve **206**, the line **207**, the valve **201**, the line **16a**, the flow path **15**, and the line **16b**. The line **207** is an example of a third line. Further, the line **16a** in the circulator **200** is provided with a temperature sensor **203** for detecting a temperature on the inlet side of the flow path **15**. The temperature sensor **203** may be disposed outside the temperature controller **20**. For example, the temperature sensor

203 may be disposed directly below the lower electrode LE, e.g., at the connection portion between the line **16a** and the flow path **15**, or may be disposed at an intermediate portion between the lower electrode LE and the temperature controller **20**.

(37) The first temperature controller **220** is connected to the line **16a** through a line **229**, a line **210**, and the valve **201**. Further, the first temperature controller **220** is connected to the line **16b** through a line **230**, a line **212**, and the check valve **204**. A connection position B between the line **229** and the line **210** and a connection position C between the line **230** and the line **212** are connected by a line **211** that is a bypass line. A pressure sensor **208** for monitoring is disposed at the connection position C.

(38) In the present embodiment, the first temperature controller **220** controls a temperature of the first heat medium on a high temperature side. The temperature-controlled first heat medium of the first temperature controller **220** passes through the line **229**, the line **210**, and the valve **201** and is mixed with the heat medium circulating from the line **207** to the line **16a**, and the mixed heat medium is supplied into the flow path **15** of the lower electrode LE. The temperature of the first heat medium may be, e.g., 90° C. The pressures in the lines **210** to **212**, **229**, and **230** are reduced by supplying the first heat medium into the flow path **15** of the lower electrode LE. The heat medium discharged from the flow path **15** passes through the check valve **204** that is opened due to a partial pressure drop at the connection position A of the line **16b**, and is returned to the temperature controller **220** through the lines **212** and **230**. The line including the lines **229**, **210**, and **16a** is an example of a supply line or a first supply line. The line including the lines **16b**, **212**, and **230** is an example of a return line or a first return line. The line including the lines **210** to **212**, **229**, and **230** is an example of a first line.

(39) The first temperature controller **220** supplies a heat medium in a reservoir tank **221** to the line **229** using a pump **222**. A heat exchanger **223**, a flow rate sensor **224**, a pressure sensor **225**, a temperature sensor **226**, and a variable valve **227** are disposed on the outlet side of the pump **222**. In other words, the heat exchanger **223**, the flow sensor **224**, the pressure sensor **225**, the temperature sensor **226**, and the variable valve **227** are disposed immediately after the pump **222**. The heat exchanger **223** heats or cools the heat medium supplied to the line **229** to a set temperature. The flow rate sensor **224** detects a flow rate of the heat medium supplied by the pump **222** on a discharge side of the line **229**. The pressure sensor **225** detects a pressure of the heat medium supplied by the pump **222** on the discharge side of the line **229**. The temperature sensor **226** detects a temperature of the heat medium supplied by the pump **222** on the discharge side of the line **229**. The variable valve **227** adjusts a pressure of the heat medium supplied by the pump **222** in the line **229** together with a variable valve **228** on the line **230** side that is the return line.

(40) The second temperature controller **240** is connected to the line **16a** through a line **249**, a line **213**, and the valve **201**. Further, the second temperature controller **240** is connected to the line **16b** through a line **250**, a line **215**, and a check valve **205**. A connection position D between the line **249** and the line **213** and a connection position E between the line **250** and the line **215** are connected by a line **214** that is a bypass line. A pressure sensor **209** for monitoring is disposed at the connection position E.

(41) In the present embodiment, the second temperature controller **240** controls a temperature of the second heat medium on a low temperature side. The temperature-controlled second heat medium of the second temperature controller **240** passes through the line **249**, the line **213**, and the valve **201** and is mixed with the heat medium circulating from the line **207** to the line **16a**, and the mixed heat medium is supplied into the flow path **15** of the lower electrode LE. The temperature of the second heat medium may be, e.g., -10° C. The pressures in the lines **213** to **215**, **249**, and **250** are reduced by supplying the second heat medium into the flow path **15** of the lower electrode LE. The heat medium discharged from the flow path **15** passes through the check valve **205** that is opened due to a partial pressure drop at the connection position A of the line **16b**, and is returned to the temperature controller **240** through the lines **215** and **250**. The line including the lines **249**, **213**,

and **16a** is an example of a supply line or a second supply line. The line including the lines **16b**, **215**, and **250** is an example of a return line or a second return line. The line including the lines **213** to **215**, **249**, and **250** is an example of a second line. The check valve (a first check valve) **204**, the check valve (a second check valve) **205** and the check valve (a third check valve) **206** constitute a check valve mechanism. The check valve mechanism is connected to the outlet of the flow path **15** and configured to branch the heat medium in three directions.

(42) The second temperature controller **240** supplies a heat medium in a reservoir tank **241** to the line **249** using a pump **242**. A heat exchanger **243**, a flow sensor **244**, a pressure sensor **245**, a temperature sensor **246**, and a variable valve **247** are disposed on the outlet side of the pump **242**. In other words, the heat exchanger **243**, the flow sensor **244**, the pressure sensor **245**, the temperature sensor **246**, and the variable valve **247** are disposed immediately after the pump **242**. The heat exchanger **243** heats or cools the heat medium supplied to the line **249** to a set temperature. The flow rate sensor **244** detects a flow rate of the heat medium supplied by the pump **242** on the discharge side of the line **249**. The pressure sensor **245** detects a pressure of the heat medium supplied by the pump **242** on the discharge side of the line **249**. The temperature sensor **246** detects a temperature of the heat medium supplied by the pump **242** on the discharge side of the line **249**. The variable valve **247** adjusts a pressure of the heat medium supplied by the pump **242** in the line **249** together with a variable valve **248** on the line **250** side that is the return line.

(43) The reservoir tank **221** of the first temperature controller **220** and the reservoir tank **241** of the second temperature controller **240** are connected by lines **251** and **252**. The lines **251** and **252** are examples of a communication line, and adjust a liquid level of the reservoir tank **221** storing the first heat medium and a liquid level of the reservoir tank **241** storing the second heat medium. The line **251** is used for movement of a gas (air, evaporated heat medium, or the like) between the reservoir tanks **221** and **241**. The line **252** is used for movement of a liquid (heat medium) between the reservoir tanks **221** and **241**. Accordingly, leakage of the heat medium is prevented.

(44) The controller **11** controls opening degrees of the valves **201**, **227**, **228**, **247**, and **248**, discharge pressures of the pumps **202**, **222**, and **242**, and temperatures in the heat exchangers **223** and **243**. The pumps **202**, **222**, and **242** can control the discharge pressure depending on an inverter frequency. The opening degree of the valve **201** can be adjusted within a range of +100% and -100%, for example. At a position where the opening degree of the valve **201** is 0%, the entire heat medium from the line **207** flows into the line **16a**, and the heat media from the lines **210** and **213** do not flow into the line **16a**. At a position where the opening degree of the valve **201** is +100%, the entire heat medium from the line **210** flows into the line **16a**, and the heat media from the lines **207** and **213** do not flow into the line **16a**. On the other hand, at a position where the opening degree of the valve **201** is -100%, the entire heat medium from the line **213** flows into the line **16a**, and the heat media from the lines **207** and **210** do not flow into the line **16a**.

(45) In other words, when it is desired to increase the temperature of the circulating heat medium obtained by adding the first heat medium on the high temperature side thereto, the opening degree of the valve **201** is changed from the 0% position to a plus side. On the other hand, when it is desired to decrease the temperature of the circulating heat medium obtained by adding the second heat medium on the low temperature side thereto, the opening degree of the valve **201** is changed from the 0% position to a minus side. For example, when plasma is ignited in the processing space PS to perform plasma processing, heat is inputted from the plasma to the lower electrode LE. Therefore, in order to keep the temperature of the lower electrode LE at a constant level, the second heat medium on the low temperature side of the line **213** is added thereto at a ratio of 10% by setting the opening degree of the valve **201** to -10% and setting the flow rate of the heat medium circulating in the line **207** to 90%, for example.

(46) (Pressure Control Method)

(47) Next, a pressure control method according to the present embodiment will be described. FIG. 3 is a flowchart showing an example of a pressure control process in the present embodiment.

(48) The controller **11** operates the pump **202** of the circulator **200** to start the circulation of the heat medium in the order of the line **16b**, the check valve **206**, the line **207**, the valve **201**, the line **16a**, the flow path **15**, and the line **16b**. The controller **11** controls the opening degree of the valve **201** based on the temperature detected by the temperature sensor **203**, and controls the temperature of the circulating heat medium to the set temperature.

(49) The controller **11** controls the pressure sensors **225** and **245** to detect the pressures in the lines **229** and **249**, respectively (step **S1**). The controller **11** compares the detected pressures in the lines **229** and **249** with set values (step **S2**). The controller **11** controls inverter frequencies of the pumps **222** and **242** such that the pressures in the lines **229** and **249** become close to the set values based on the comparison result (step **S3**). In other words, the controller **11** controls the discharge pressures of the pump **222** and **242** based on the difference between the pressures detected by the pressure sensors **225** and **245** and the set values of the discharge pressures of the pump **222** and **242**. Accordingly, it is possible to improve the temperature controllability for the control target (the lower electrode **LE**).

(50) (Operation of Temperature Controller **20**)

(51) Next, an operation of the temperature controller **20** in the case where the temperature detected by the temperature sensor **203** is lower than 35° C. that is the set temperature of the circulating heat medium, i.e., the set temperature of the lower electrode **LE** as a control target, and an operation of the temperature controller **20** in the case where the temperature detected by the temperature sensor **203** is higher than 35° C. will be described. In the following description, the first set value of the pressure in the line **229** in the first temperature controller **220** and the second set value of the pressure in the line **249** in the second temperature controller **240** on the low temperature side, i.e., the set values of the discharge pressures of the pumps **222** and **242**, are set to 0.7 MPa to 1.0 MPa. Further, the check valve **204** connected to the line **212** on the high temperature side is adjusted to be opened at 0.580 MPa to 0.590 MPa, and the check valve **205** connected to the line **215** on the low temperature side is adjusted to be opened at 0.570 to MPa 0.580 MPa. In other words, the pressure difference between the high temperature side and the low temperature side is set to 0.010 MPa (high temperature side>low temperature side). The above-described set values of the pressures are examples, and the set values of the pressures are not limited thereto.

(52) First, when the temperature detected by the temperature sensor **203** is lower than the set temperature, the controller **11** changes the opening degree of the valve **201** from the current position to the plus side so that the first heat medium from the line **210** flows into the line **16a**. When the first heat medium from the line **210** flows into the line **16a**, the pressure in the line **210** is reduced, and the pressures in the lines **211**, **212**, **229**, and **230** connected to the line **210** are also reduced. In other words, the pressures detected by the pressure sensors **208** and **225** are reduced.

(53) Before the opening degree of the valve **201** is changed, the second heat medium on the low temperature side flows into the line **16a**. In this case, if the position of the opening degree of the valve **201** is changed to the plus side, the line **213** side of the valve **201** is closed, so that the pressure in the line **215** on the low temperature side increases and the check valve **205** connected to the line **215** is changed from the open state to the closed state. On the other hand, on the high temperature side, the pressure in the line **212** is reduced, so that the check valve **204** is opened and the heat medium corresponding to the inflow amount flows from the line **16b** to the line **212**.

(54) The controller **11** compares the pressure in the line **229** detected by the pressure sensor **225** with the first set value, and increases the discharge pressure by controlling the inverter frequency of the pump **222** such that the reduced pressure in the line **229** becomes close to the set value. In other words, feedback control is performed based on the pressure in the line **229**.

(55) When the discharge pressure of the pump **222** increases, the pressure in the line **229** increases and the pressures in the lines **210** to **212** and **230** connected to the line **229** also increase. When the temperature detected by the temperature sensor **203** reaches the set temperature, the controller **11** changes the opening degree of the valve **201** from the current position to the minus side. When the

opening of the valve **201** is changed from the current position to the minus side, the pressure in the line **210** increases and the pressures in the lines **211**, **212**, **229**, and **230** connected to the line **210** also increase. Accordingly, the check valve **204** is closed.

(56) Next, when the temperature detected by the temperature sensor **203** is higher than the set temperature, the controller changes the opening degree of the valve **201** from the current position to the minus side so that the second heat medium from the line **213** flows into the line **16a**. When the second heat medium from the line **213** flows into the line **16a**, the pressure in the line **213** is reduced and the pressures in the lines **214**, **215**, **249**, and **250** connected to the line **213** are also reduced. In other words, the pressures detected by the pressure sensors **209** and **245** are reduced.

(57) Before the opening degree of the valve **201** is changed, the first heat medium on the high temperature side flows into the line **16a**. In this case, if the position of the opening degree of the valve **201** is changed to the minus side, the line **210** side of the valve **201** is closed, so that the pressure in the line **212** on the high temperature side increases and the check valve **204** connected to the line **212** is changed from an open state to a closed state. On the other hand, on the low temperature side, the pressure in the line **215** is reduced, so that the check valve **205** is opened and a heat medium corresponding to the inflow amount flows from the line **16b** to the line **215**.

(58) The controller **11** compares the pressure in the line **249** detected by the pressure sensor **245** with the second set value, and increases the discharge pressure by controlling the inverter frequency of the pump **242** such that the reduced pressure in the line **249** becomes close to the set value. In other words, the feedback control is performed based on the pressure in the line **249**.

(59) When the discharge pressure of the pump **242** increases, the pressure in the line **249** increases and the pressures in the lines **213** to **215**, and **250** connected to the line **249** also increase. When the temperature detected by the temperature sensor **203** reaches the set temperature, the controller **11** changes the opening degree of the valve **201** from the current position to the plus side. When the opening of the valve **201** is changed from the current position to the plus side, the pressure in the line **213** increases and the pressures in the lines **214**, **215**, **249**, and **250** connected to the line **213** also increase, so that the check valve **205** is closed. In this manner, the time in which the check valves **204** and **205** are opened simultaneously can be shortened, which makes it possible to improve the temperature controllability for the control target (the lower electrode LE). In other words, since the pressure feedback control is performed depending on the pressure difference between the check valves **204** and **205**, the responsiveness is improved and the pressure variation can be suppressed. The temperature controller **20** may include a control device for performing control based on information from the controller **11**. In this case, the feedback control may be performed based on the pressures detected by the pressure sensors **208** and **209** instead of the pressures detected by the pressure sensors **225** and **245**.

(60) (Test Results)

(61) Next, temperature characteristics in a test example related to the present embodiment and temperature characteristics in a comparative example in which the inverter frequencies of the pumps **222** and **242** are fixed will be described with reference to FIGS. **4** and **5**. FIG. **4** shows an example of the temperature characteristics in the test example. In a graph **300** of FIG. **4**, a graph **301** showing the set temperature of the first heat medium, a graph **302** of the temperature sensor **226** showing the heat medium temperature on the discharge side, and a graph **303** showing the heat medium temperature on the return side of the line are illustrated as graphs on the high temperature side. Further, in the graph **300**, a graph **304** showing the set temperature of the second heat medium, a graph **305** of the temperature sensor **246** showing the heat medium temperature on the discharge, and a graph **306** showing the heat medium temperature on the return side of the line are illustrated as graphs on the low temperature side. Further, in the graph **300**, a graph **307** shows the opening degree of the valve **201**. The vertical axis of the graph **300** represents both a temperature and an opening degree.

(62) In the graph **300**, the temperature characteristics in an extreme example in which the set

temperature of the control target is changed from a low temperature (10° C.) to a high temperature (70° C.) and then changed from the high temperature (70° C.) to the low temperature (10° C.) are illustrated. The first heat medium is set to 90° C. as shown in the graph **301**, and the second heat medium is set to -10° C. as shown in the graph **304**.

(63) When the set temperature of the control target is changed from 10° C. to 70° C. at a timing **308**, the opening degree of the valve **201** changes considerably to the plus side as shown in the graph **307** and, thus, the first heat medium flows into the flow path of the control target. As shown in the graph **302**, the temperature of the first heat medium on the return side decreases to 39.2° C. immediately after the timing **308**, but returns to a temperature slightly lower than 90° C. at a timing **309** when the first heat medium that has flown into the flow path of the control target reaches the return side. Similarly, as shown in the graph **302**, the temperature on the discharge side of the first heat medium decreases after the timing **308**, but substantially returns to the set temperature of 90° C. at the timing **309**.

(64) Next, when the set temperature of the control target is changed from 70° C. to 10° C. at the timing **309**, the opening degree of the valve **201** changes considerably to the minus side as shown in the graph **307** (partially omitted in FIG. 4), so that the second heat medium flows into the flow path of the control target. Further, as shown in the graph **306**, the temperature of the second heat medium on the return side increases up to 44.0° C. immediately after the timing **309**, but returns to about -5° C. at a timing corresponding to the right end of the graph **300** when the second heat medium that has flown into the flow path of the control target reaches the return side. Similarly, as shown in the graph **305**, the temperature on the discharge side of the second heat medium increases after the timing **309**, but returns to the set temperature of -10° C. at the timing corresponding to the right end of the graph **300**.

(65) FIG. 5 shows an example of temperature characteristics in the comparative example. In a graph **320** of FIG. 5, a graph **321** showing the set temperature of the first heat medium, a graph **322** showing the heat medium temperature on the discharge side, and a graph **323** showing the heat medium temperature on the return side of the line are illustrated as graphs on the high temperature side. Further, in the graph **320**, a graph **324** showing the set temperature of the second heat medium, a graph **325** showing the heat medium temperature on the discharge side, and a graph **326** showing the heat medium temperature on the return side of the line are illustrated as graphs on the low temperature side. Further, in the graph **320**, a graph **327** shows the opening degree of the valve **201**. The vertical axis of the graph **320** represents both a temperature and an opening degree.

(66) In the graph **320**, similarly to the graph **300** of FIG. 4, the temperature characteristics in the case where the set temperature of the control target is changed from the low temperature (10° C.) to the high temperature (70° C.) and then changed from the high temperature (70° C.) to the low temperature (10° C.) are illustrated. The first heat medium is set to 90° C. as shown in the graph **321**, and the second heat medium is set to -10° C. as shown in the graph **324**.

(67) When the set temperature of the control target is changed from 10° C. to 70° C. at a timing **328**, the opening degree of the valve **201** changes considerably to the plus side as shown in the graph **327**, so that the first heat medium flows into the flow path of the control target. Further, as shown in the graph **323**, the temperature of the first heat medium on the return side decreases to 52.7° C. immediately after the timing **328**, but returns to about 80° C. when the first heat medium that has flown into the flow path of the control target reaches the return side. However, even at the timing **329**, the temperature is slightly lower than 85° C. without returning to the set temperature of 90° C. Similarly, as shown in the graph **322**, the temperature on the discharge side of the first heat medium decreases after the timing **328** and reaches about 80° C. without returning to the set temperature of 90° C. even at the timing **329**. On the other hand, as shown in the graph **326**, the temperature of the second heat medium on the return side increases to about 0° C. immediately after the timing **328**, so that it is considered that the check valve **205** on the second heat medium side and the check valve **204** on the first heat medium side are opened simultaneously. It is

considered that this is because the temperature on the return side and the temperature on the discharge side of the first heat medium did not return to the level close to the set temperature even at the timing **329**.

(68) Next, when the set temperature of the control target is changed from 70° C. to 10° C. at the timing **329**, as shown in the graph **327**, the opening degree of the valve **201** changes considerably to the minus side (partially omitted in FIG. 5), so that the second heat medium flows into the flow path of the control target. Further, as shown in the graph **326**, the temperature of the second heat medium on the return side increases to about 28.9° C. immediately after the timing **329**, but returns to about -5° C. at the timing corresponding to the right end of the graph **320** when the second heat medium that has flow into the flow path of the control target reaches the return side. Similarly, as shown in the graph **325**, the temperature on the discharge side of the second heat medium increases after the timing **329**, but returns to the set temperature of -10° C. at the timing corresponding to the right end of the graph **300**.

(69) According to the test results of the graphs **300** and **320**, in the comparative example, when the temperature is changed from low temperature to high temperature, the temperature of the second heat medium on the return side is maintained at a high level, so that the check valve **205** is opened. On the other hand, in the test example, the temperature of the second heat medium on the return side hardly changes, so that the check valve **205** is closed. In other words, it is clear that the temperature controllability is improved in the test example than in the comparative examples.

First Modified Example

(70) In the above embodiment, the temperature controller **20** of the type in which the heat medium circulates in the circulator **200** has been described. However, it is also possible to use a temperature controller of a type in which the heat medium flowing through the flow path **15** in the lower electrode LE is switched between the first heat medium and the second heat medium. Such an embodiment will be described as a first modified example. Since the partial configuration of the substrate processing apparatus and the temperature controller and the pressure control method in the first modified example are the same as those in the above-described embodiment, redundant description thereof will be omitted.

(71) FIG. **6** shows an example of the temperature controller in the first modified example. A temperature controller **20a** shown in FIG. **6** is different from the temperature controller **20** in the above embodiment in that it includes a switching device **260**, a first temperature controller **220a**, and a second temperature controller **240a**, instead of the circulator **200**, the first temperature controller **220**, and the second temperature controller **240**. The switching device **260** has valves **261** to **264**, temperature sensors **265**, and lines **266** to **271**.

(72) The first temperature controller **220a** is connected to the line **16a** through a line **272**, the line **266**, and the valve **261**. Further, the first temperature controller **220a** is connected to the line **16b** through a line **273**, the line **268**, and the valve **262**. In the first modified example, the first temperature controller **220a** controls the temperature of the first heat medium on the high temperature side. The first temperature controller **220a** supplies the temperature-controlled first heat medium into the flow path **15** of the lower electrode LE through the line **272**, the line **266**, the valve **261**, and the line **16a**. The heat medium supplied into the flow path **15** of the lower electrode LE is returned to the first temperature controller **220a** through the line **16b**, the valve **262**, the line **268**, and the line **273**. The line including the lines **272** and **266** and the line **16a** is an example of the supply line or the first supply line. Further, the line including the lines **16b**, **268**, and **273** is an example of the return line or the first return line.

(73) The second temperature controller **240a** is connected to the line **16a** through a line **274**, the line **269**, and the valve **261**. Further, the second temperature controller **240a** is connected to the line **16b** through a line **275**, the line **271**, and the valve **262**. In the first modified example, the second temperature controller **240a** controls the temperature of the second heat medium on the low temperature side. The second temperature controller **240a** supplies the temperature-controlled

second heat medium into the flow path **15** of the lower electrode LE through the line **274**, the line **269**, the valve **261**, and the line **16a**. The heat medium supplied into the flow path **15** of the lower electrode LE is returned to the second temperature controller **240a** through the line **16b**, the valve **262**, the line **271**, and the line **275**. The line including the line **274**, the line **269**, and the line **16a** is an example of the supply line or the second supply line. Further, the line including the line **16b**, the line **271**, and the line **275** is an example of the return line or the second return line.

(74) The valve **261** is disposed at the connection portion of the line **16a**, the line **266**, and the line **269**, and switches the heat medium flowing in the flow path **15** of the lower electrode LE to the first heat medium or the second heat medium. The valve **262** is disposed at the connection portion of the line **16b**, the line **268**, and the line **271**, and switches an output destination of the heat medium discharged from the flow path **15** of the lower electrode LE to the first temperature controller **220a** or the second the temperature controller **240a**.

(75) A connection position F between the line **272** and the line **266** and a connection position G between the line **273** and the line **268** are connected by the line **267** that is a bypass line. The line **267** is provided with a bypass valve **263**.

(76) A connection position H between the line **274** and the line **269** and a connection position I between the line **275** and the line **271** are connected by the line **270** that is a bypass line. The line **270** is provided with a bypass valve **264**.

(77) The line **16a** in the temperature controller **20a** is provided with the temperature sensor **265** for measuring the temperature on the inlet side of the flow path **15**. The temperature sensor **265** may be disposed outside the temperature controller **20a**. For example, the temperature sensor **265** may be disposed directly under the lower electrode LE, e.g., at the connection portion of the line **16a** and the flow path **15**, or may be disposed at an intermediate portion between the lower electrode LE and the temperature controller **20a**.

(78) The opening and closing of the valves **261** and **262** and the bypass valves **263** and **264** are controlled by the controller **11**. The first temperature controller **220a** and the second temperature controller **240a** are the same as the first temperature controller **220** and the second temperature controller **240** related to the above embodiment except that the variable valves **227** and **228**, and the variable valves **247** and **248** are omitted, so that the description thereof will be omitted.

(79) In the temperature controller **20a**, when the first heat medium on the high temperature side is supplied into the flow path **15** of the lower electrode LE, the valve **261** is opened on the line **266** side and closed on the line **269** side, and the valve **262** is opened on the line **268** side and closed on the line **271** side. The bypass valve **263** is closed and the bypass valve **264** is opened. Therefore, the first heat medium on the high temperature side supplied from the first temperature controller **220a** is supplied into the flow path **15** of the lower electrode LE and returned to the first temperature controller **220a**, and the second heat medium on the low temperature side supplied from the second temperature controller **240** is returned to the second temperature controller **240a** through the bypass valve **264**.

(80) On the other hand, in the temperature controller **20a**, when the second heat medium on the low temperature side is supplied into the flow path **15** of the lower electrode LE, the valve **261** is closed on the line **266** side and opened on the line **269** side, and the valve **262** is closed on the line **268** side and opened on the line **271** side. The bypass valve **263** is opened and the bypass valve **264** is closed. Therefore, the second heat medium on the low temperature side supplied from the second temperature controller **240a** is supplied into the flow path **15** of the lower electrode LE and returned to the second temperature controller **240a**, and the first heat medium on the high temperature side supplied from the first temperature controller **220a** is returned to the first temperature controller **220a** through the bypass valve **263**.

(81) (Pressure Control Method)

(82) The controller **11** performs a pressure control process in the temperature controller **20a** in the same manner as that shown in FIG. 3 of the above embodiment. In other words, the controller **11**

controls the pressure sensors **225** and **245** to detect the pressures in the lines **272** and **274**, respectively (step **S1**). The controller **11** compares the detected pressures in the lines **272** and **274** with the set values (step **S2**). The controller **11** controls the inverter frequencies of the pumps **222** and **242** such that the pressures in the lines **272** and **274** become close to the set values based on the comparison result (step **S3**). In other words, the controller **11** controls the discharge pressures of the pumps **222** and **242** based on the difference between the pressures detected by the pressure sensors **225** and **245** and the set values of the discharge pressures of the pumps **222** and **242**. Since the variation in the pressures in the lines **272** and **274** for supplying the heat medium is suppressed, the temperature controllability for the control target (the lower electrode **LE**) can be improved.

Second Modified Example

(83) In the above embodiment and first modified example, the temperature controllers **20** and **20a** for supplying a heat medium to one control target have been described. However, the present disclosure may be applied to a temperature controller for controlling a temperature of an upper system such as the upper electrode disposed at an upper portion of the processing chamber or the like. Such an embodiment will be described as a second modified example. Since the partial configuration of the substrate processing apparatus and the temperature controller and the pressure control method in the second modified example are the same as those in the above embodiment and first modified example, redundant description thereof will be omitted.

(84) FIG. **7** shows an example of a control target in the second modified example. A substrate processing apparatus **400** shown in FIG. **7** is an example of a substrate processing apparatus having an upper electrode provided with a cooling plate **410** as a control target at an upper portion of a processing chamber **401**. In the substrate processing apparatus **400**, the cooling plate **410** is supported at the upper portion of the processing chamber **401** via a support member. The cooling plate **410** has a heater **411** and an electrode plate **412**. The electrode plate **412** that is an example of the upper electrode has a bottom surface facing the processing space and faces an upper surface of a substrate support **413** having a lower electrode. A ceiling plate **402** is supported by the processing chamber **401** at a position above the cooling plate **410**. In FIG. **7**, a system for supplying a processing gas supply and a system for supplying an RF power to the upper electrode are omitted.

(85) The cooling plate **410** has therein a flow path **409** through which a heat medium flows. A flow path **408** through which a heat medium flows is formed at an upper portion of the processing chamber **401**. Similarly, the ceiling plate **402** has therein a flow path **406** through which a heat medium flows. The flow paths **406** and **408** are connected by a flow path **407**. In other words, the flow paths **406** to **408** for controlling the temperatures of the processing chamber **401** and the ceiling plate **402**, and flow paths **409** for controlling the temperature of the cooling plate **410** are formed at the upper portion of the processing chamber **401**. The cooling plate **410** is cooled by increasing the flow rate of the heat medium of, e.g., 35° C., when heat is inputted from the plasma, and is adjusted to, e.g., 150° C., by decreasing the flow rate of the heat medium and performing heating using the heater **411** in an idle state.

(86) A temperature controller **20b** supplies a heat medium from a line **403** connected to the flow path **406** into the flow paths **406** to **408**. The heat medium supplied into the flow paths **406** to **408** is returned to the temperature controller **20b** through a line **404** connected to the flow path **408**. Further, the temperature controller **20b** supplies the heat medium into the flow path **409** through the line **403** and a flow rate control valve **405**. The heat medium supplied into the flow path **409** is returned to the temperature controller **20b** through the line **404** connected to the flow path **409**.

(87) FIG. **8** shows an example of the temperature controller in the second modified example. The temperature controller **20b** shown in FIG. **8** is different from the temperature controller **20** in the above embodiment in that it does not have the circulator **200** and has a controller **240b** instead of the first temperature controller **220** and the second temperature controller **240**. In other words, the temperature controller **20b** has the temperature controller **240b** for supplying a heat medium controlled to one set temperature, instead of multiple temperature controllers on a high temperature

side and a low temperature side. In FIG. 8, the path for supplying the heat medium into the flow paths **406** to **408** is omitted. Further, the temperature controller **20b** has therein a controller **280**. The controller **280** controls individual components in the temperature controller **20b** based on the information from the substrate processing apparatus **400**.

(88) The temperature controller **240b** is connected to a plurality of substrate processing apparatuses **400** through the lines **403** and **404**. In FIG. 8, two substrate processing apparatuses **400** and **400a** are illustrated, and flow rate control valves **405** and **405a** corresponding to the substrate processing apparatuses **400** and **400a** are illustrated. In FIG. 8, the line **403** is branched into lines **422** and **422a** corresponding to the substrate processing apparatuses **400** and **400a**. Further, lines **423** and **423a** corresponding to the substrate processing apparatuses **400** and **400a** are joined to the line **404**.

(89) The temperature controller **240b** is connected to the substrate processing apparatus **400** through the line **403**, the line **421**, the flow rate control valve **405**, and the line **422**. Further, the temperature controller **240b** is connected to the substrate processing apparatus **400** through the line **404**, the line **424**, and the line **423**. In the second modified example, the temperature controller **240b** controls the temperature of the heat medium set to the set temperature (e.g., 35° C.). When the flow rate control valve **405** is opened, the temperature controller **240b** supplies the temperature-controlled heat medium into the flow path **409** of the substrate processing apparatus **400** through the line **403**, the line **421**, the flow rate control valve **405**, and the line **422**. The heat medium supplied into the flow path **409** is returned to the temperature controller **240b** through the line **423**, the line **424**, and the line **404**. The line including the line **403**, the line **421**, and the line **422** is an example of the supply line. The line including of the line **423**, the line **424**, and the line **404** is an example of the return line.

(90) Similarly, the temperature controller **240b** is connected to the substrate processing apparatus **400a** through the line **403**, the line **421**, the flow rate control valve **405a**, and the line **422a**. Further, the temperature controller **240b** is connected to the substrate processing apparatus **400a** through the line **404**, the line **424**, and the line **423a**. When the flow rate control valve **405a** is opened, the temperature controller **240b** supplies the temperature-controlled heat medium into the flow path **409** of the substrate processing apparatus **400a** through the line **403**, the line **421**, the flow rate control valve **405a**, and the line **422a**. The heat medium supplied into the flow path **409** is returned to the temperature controller **240b** through the line **423a**, the line **424**, and the line **404**.

(91) A connection position J between the line **403** and the line **421** and a connection position K between the line **404** and the line **424** are connected by a line **425** that is a bypass line. The line **425** is provided with a bypass valve **420**. The bypass valve **420** is opened when both the flow control valves **405** and **405a** at the downstream side are closed, so that the heat medium flowing in the flow path **409** is returned from the line **403** to the line **404**.

(92) The opening and closing of the flow rate control valves **405** and **405a** and the bypass valve **420** are controlled by the controller **280**. Since the temperature controller **240b** is the same as the second temperature controller **240** related to the above embodiment except that the variable valves **247** and **248** are omitted, the description thereof will be omitted.

(93) Since the flow rate control valves **405** and **405a** are opened and closed depending on the temperature control in the substrate processing apparatuses **400** and **400a**, the pressure in the line **403** on the discharge side of the heat medium varies depending on the opening and closing of the flow rate control valves **405** and **405a**.

(94) (Pressure Control Method)

(95) The controller **280** performs the pressure control process in the same manner as that shown in FIG. 3 of the above embodiment. In other words, the controller **280** controls the pressure sensor **245** to detect the pressure in the line **403** (step S1). The controller **280** compares the detected pressure in the line **403** with the set value (step S2). The controller **280** controls the inverter frequency of the pump **242** such that the pressure in the line **403** becomes close to the set value based on the comparison result (step S3). In other words, the controller **280** controls the discharge

pressure of the pump **242** based on the difference between the pressure detected by the pressure sensor **245** and the set value of the discharge pressure of the pump **242**. Since the variation of the pressure in the line for supplying the heat medium is suppressed, the temperature controllability for the control target (the cooling plate **410**) can be improved.

(96) As described above, in accordance with the present embodiment and the modified examples, the temperature controllers **20**, **20a**, and **20b** includes the supply lines (the lines **229**, **249**, **272**, **274**, and **403**) for supplying the heat medium to the control target (the lower electrode LE and the cooling plate **410**), the return lines (the lines **230**, **250**, **273**, **275**, and **404**) for returning the heat medium from the control target, the pumps **222** and **242** for discharging the heat medium to the supply lines, the pressure sensors **225** and **245** for detecting the pressures in the supply lines immediately after the pumps, and the controller **280**. The controller controls the discharge pressures of the pumps based on the difference between the pressures detected by the pressure sensors and the set values of the discharge pressures of the pumps. Accordingly, the temperature controllability for the control target can be improved. The feedback control may be performed based on the pressures detected by the pressure sensors **208** and **209** instead of the pressures detected by the pressure sensors **225** and **245**. Further, the feedback control may be performed based on the pressures detected by the pressure sensors additionally disposed at the supply lines (the lines **229**, **249**, **272**, **274**, and **403**) for supplying the heat medium.

(97) Further, in accordance with the present embodiment and the modified examples, the discharge pressure is controlled by controlling the inverter frequency of the pump. Accordingly, it is possible to suppress variation of the pressure in the line for supplying the heat medium.

(98) Further, in accordance with the second modified example, the number of control targets is two or more. Accordingly, even when the pressure in the line for supplying the heat medium varies depending on the operations of the multiple control targets, it is possible to suppress the variation of the pressure in the line for supplying the heat medium.

(99) Further, in accordance with the present embodiment, the substrate processing apparatus **1** includes the substrate support PD disposed in the processing chamber **12** to place the substrate (wafer W) thereon, the first temperature controller **220** configured to supply the heat medium having the first temperature to the flow path **15** disposed in the substrate support PD through the first line (the lines **229**, **210**, and **16a**), the second temperature controller **240** configured to supply the heat medium having the second temperature to the flow path **15** through the second line (the lines **249**, **213**, and **16a**), the third line (the lines **16b**, **207**, and **16a**) configured to circulate the heat medium while being connected from the outlet of the flow path **15** to the inlet of the flow path **15**, a check valve mechanism connected to the outlet of the flow path **15** and configured to branch the heat medium in three directions, the check valve mechanism including the first check valve **204** connected to the first line, the second check valve **205** connected to the second line, and the third check valve **206** that is connected to the third line and opened when the heat medium circulates, and the control device (the controller **11**). The controller detects the pressures in the first line and the second line. Further, the controller compares the detected pressure in the first line with the first set value of the pressure in the first line, and compares the detected pressure in the second line with the second set value of the pressure in the second line. Moreover, when the first check valve is opened due to the pressure difference between the outlet side of the flow path **15** and the first line, the controller controls the first temperature controller **220** such that the pressure in the first line becomes close to the first set value of the pressure in the first line based on the comparison result. When the second check valve is opened due to the pressure difference between the outlet side of the flow path **15** and the second line, the controller controls the second temperature controller **240** such that the pressure in the second line becomes close to the second set value based on the comparison result. Accordingly, the temperature controllability for the control target (the lower electrode LE) can be improved by controlling the pressures on the discharge sides of the pumps **222** and **242**) to be constant.

(100) Further, in accordance with the present embodiment, the valve **201** for controlling a mixing ratio of the heat media flowing through the first line, the second line, and the third line, and the temperature sensor **203** for detecting the temperature of the mixed heat medium are disposed on the inlet side of the flow path **15**. The opening degree of the valve **201** is controlled such that the heat medium flowing through the first line or the second line are mixed with the heat medium flowing through the third line depending on the temperature detected by the temperature sensor **203**. Accordingly, the temperature controllability for the control target (the lower electrode LE) can be improved.

(101) Further, in accordance with the present embodiment, the pressure in the first line is detected by the first pressure sensor (the pressure sensor **225**) disposed immediately after the pump **222** of the first temperature controller **220**, and the pressure in the second line is detected by the second pressure sensor (the pressure sensor **245**) disposed immediately after the pump **242** of the second temperature controller **240**. Accordingly, the pressures of the lines immediately after the pumps can be feedback-controlled by the inverter frequencies of the pumps.

(102) Further, in accordance with the present embodiment, the discharge pressures of the pumps are controlled by controlling the inverter frequencies of the pumps **222** and **242**. Accordingly, the discharge pressures of the pumps can be stabilized.

(103) The embodiments of the present disclosure are illustrative in all respects and are not restrictive. The above-described embodiments may be omitted, replaced, or changed in various forms without departing from the scope of the appended claims and the gist thereof.

(104) Although capacitively coupled plasma (CCP) is used as an example of a plasma source in the above embodiment, the present disclosure is not limited thereto. The plasma source may be, e.g., inductively coupled plasma (ICP), microwave excited surface wave plasma (SWP), electron cyclotron resonance plasma (ECP), helicon wave plasma (HWP), or the like.

(105) Although the plasma etching processing apparatus has been described as an example of the substrate processing apparatus **1** in the above embodiment, the present disclosure is not limited thereto. The present disclosure may be applied to a film forming apparatus, a modifying apparatus, a cleaning device, or the like, other than an etching apparatus, as long as the apparatus controls a temperature of a temperature control target such as the wafer W or the like using a temperature-controlled heat medium.

(106) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A temperature controller comprising: a supply line configured to supply a heat medium to at least one control target; a return line configured to return the heat medium from control target; a first pump configured to discharge the heat medium to the supply line; a first pressure sensor configured to detect a first pressure in the supply line disposed immediately after the first pump; a second pump configured to discharge the heat medium to the supply line; a second pressure sensor configured to detect a second pressure in the supply line disposed immediately after the second pump; a temperature sensor configured to detect a temperature in the supply line; a valve configured to control a mixing ratio of the heat medium discharged by the first pump and the heat medium discharged by the second pump; and a controller, wherein the controller is configured to: when the temperature detected in the supply line is lower than a set temperature, control a

discharge pressure of the first pump based upon a difference between the first pressure detected by the first pressure sensor and a first set value of the discharge pressure of the first pump, when the temperature detected in the supply line reaches the set temperature control an opening degree of the valve configured to reduce a contribution of the heat medium supplied by the first pump, and when the temperature detected in the supply line is higher than the set temperature, control a discharge pressure of the second pump based upon a difference between the second pressure detected by the second pressure sensor and a second set value of the second pump.

2. The temperature controller of claim 1, wherein the controller is configured to: control the discharge pressure of the first pump by controlling an inverter frequency of the first pump, and control the discharge pressure of the second pump by controlling an inverter frequency of the second pump.

3. The temperature controller of claim 1, wherein the number of the control target is two or more.

4. The temperature controller of claim 2, wherein the number of the control target is two or more.

5. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is lower than the set temperature, increase the discharge pressure of the first pump such that the first pressure detected by the first pressure sensor comes close to the first set value.

6. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature, increase the discharge pressure of the second pump such that the second pressure detected by the second pressure sensor comes close to the second set value.

7. The temperature controller of claim 5, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature, increase the discharge pressure of the second pump such that the second pressure detected by the second pressure sensor comes close to the second set value.

8. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is lower than the set temperature, control the opening degree of the valve to increase the contribution of the heat medium supplied by the first pump.

9. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature, control the opening degree of the valve to increase the contribution of the heat medium supplied by the second pump.

10. The temperature controller of claim 8, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature, control the opening degree of the valve to increase the contribution of the heat medium supplied by the second pump.

11. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is lower than the set temperature: control the opening degree of the valve to increase the contribution of the heat medium supplied by the first pump, and increase the discharge pressure of the first pump such that the first pressure detected by the first pressure sensor comes close to the first set value.

12. The temperature controller of claim 1, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature: control the opening degree of the valve to increase the contribution of the heat medium supplied by the second pump, and increase the discharge pressure of the second pump such that the second pressure detected by the second pressure sensor comes close to the second set value.

13. The temperature controller of claim 11, wherein the controller is configured to, when the temperature detected in the supply line is higher than the set temperature: control the opening degree of the valve to increase the contribution of the heat medium supplied by the second pump, and increase the discharge pressure of the second pump such that the second pressure detected by the second pressure sensor comes close to the second set value.
